

Teacher Guide — senior / module-01- what-is-iks

IKS Curriculum

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Overview

Module 1 — What is IKS? Critical Survey and the Policy Question

Age band: Senior · **Grades:** 9–12 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Articulate** what “Indian Knowledge Systems” refers to and what it doesn’t, with awareness that the term is itself a recent (post-2000) academic and policy construction.
2. **Enumerate** the canonical structure: 4 Vedas × 4 layers, 6 Vedangas, 4 Upavedas, 6 darśanas (philosophical schools), plus modern syntheses.
3. **Read** at least one verse from a Vedic source (Ṛg Veda 1.1) and one Upanishadic verse (Īśa Upaniṣad opening) in IAST with translation.
4. **Compare** the Indian intellectual tradition with one other major tradition (Greek, Chinese, Arab-Islamic) on a specific point of inquiry.
5. **Engage** the contemporary policy debate (NEP 2020 §4.6) about integrating IKS into mainstream education — articulating positions for, against, and qualified.
6. **Construct** a 5–6 page research paper on a specific IKS claim, citing primary, secondary scholarly, and comparative sources.

NEP 2020 alignment

- §4.6 — Integration of IKS into mainstream education (the module engages this directly).
- §22 — Indian languages, arts, and culture.
- §11.8 — Multidisciplinary linkage.

Prerequisites

- Comfort reading Sanskrit in IAST with translation.
- Familiarity with structured argumentative writing.

- Basic knowledge of world intellectual history (Greek philosophy, scientific revolution).

Key vocabulary (Senior additions)

- *darśana* — “view”; the six classical philosophical systems
- *śāstra* — “treatise”
- *jñāna-mīmāṃsā* — “investigation of knowledge”
- *Yavana* — “Greek” (Indo-Greek interactions)
- *pramāṇa* — “means of valid knowledge” (the Indian epistemological taxonomy)

Materials checklist

- Printed extracts:
 - Ṛg Veda 1.1 (IAST + translation)
 - Īśa Upaniṣad opening verse
 - One Aryabhaṭīya verse
 - One Bhāskara II *Līlāvātī* problem
- Plofker (2009) extract on Mādhava
- Pollock (2006) extract on Sanskrit transmission

Day-by-day overview

Day	Title	Focus
1	“IKS”: What does the term refer to?	Boundary problem; recent construction
2	The Vedic Corpus — close reading	Ṛg 1.1 + Īśa Upaniṣad primary sources
3	The Six Darśanas	Sāṅkhya, Yoga, Nyāya, Vaiśeṣika, Mīmāṃsā, Vedānta
4	Indian Mathematics — specifics	Bakhshali, Bhāskara II, Mādhava
5	Mid-module + paper topic	Formative + outline
6	Indian Astronomy + Yavana exchange	Aryabhata, Sūrya-siddhānta, Greek-Indian interaction
7	The Policy Debate: IKS in Education	Structured debate
8	Project Session 1	Outline + thesis
9	Project Session 2	Draft + counter-argument
10	Final + paper submission	Summative + paper

Citation sources

- Primary: Ṛg Veda 1.1, Īśa Upaniṣad, Aryabhaṭīya, Bhāskara II *Līlāvātī*
- Secondary scholarly: Plofker (2009), Pollock (2006), Wujastyk (2003), Cardona (1997)
- Comparative: Lloyd & Sivin (2002) on Greek/Chinese science; Saliba (2007) on Arab-Islamic science

- Policy: NEP 2020 (Government of India, 2020); critiques by various scholars

See [sources.md](#).

Project brief

5–6 page research paper. See [assessments/project-brief.md](#).

10-Day Lesson Plan

Module 1 (Senior) — 10-Day Lesson Plan

Day	Title	File
1	“IKS”: What does the term refer to?	days/day-01-hook.md
2	Vedic corpus — close reading	days/day-02-vedic-reading.md
3	Six darśanas	days/day-03-darshanas.md
4	Indian Mathematics specifics	days/day-04-mathematics.md
5	Mid-module + paper topic	days/day-05-mid-module.md
6	Astronomy + Yavana exchange	days/day-06-astronomy.md
7	Policy debate	days/day-07-policy-debate.md
8	Project Session 1	days/day-08-project-session-1.md
9	Project Session 2	days/day-09-project-session-2.md
10	Final + paper	days/day-10-final-assessment.md

Pacing

- Day 2 needs quiet reading time — schedule when class is fresh.
- Day 7 debate can run long — book buffer.

What students leave with

- The IKS map at scholarly depth.
- Primary source reading skill (Sanskrit + IAST).

- A defensible position on the policy debate.
- One research paper.

Teacher Notes

Teacher Notes — Module 1 (Senior)

Posture

Senior students can hold this honestly. The framing:

“IKS” as a unified label is a recent (post-2000) policy construction. The intellectual traditions it groups are ancient. The label is new. The contents are real. The relationship is worth examining.

Three failure modes to push against:

1. **Triumphalism** — “Indians invented everything.” Specific claims, please.
2. **Dismissal** — “It’s all religion.” Demonstrably false at the textual level.
3. **Uncritical NEP boosterism** — “NEP §4.6 says so, therefore it must be done.” A defensible policy position is one *with* critical engagement.

On the policy debate (Day 7)

Three real positions in scholarly literature:

- **Integration (yes)** — defended by, e.g., Kapil Kapoor and others. IKS frameworks are real intellectual achievements, students benefit from knowing them, NEP §4.6 codifies this.
- **Separation (no)** — defended by, e.g., Romila Thapar, Ramachandra Guha. IKS is rich intellectual history, but mixing it with current scientific curriculum risks confusion about epistemic standards.
- **Qualified integration (yes-but)** — defended by, e.g., Manjul Bhargava-style “math is universal” view: yes to historical context, no to claims of equivalent modern validity.

Steel-man all three. The student’s job is to *articulate* each, not to pick “the right one.”

On primary source reading (Day 2)

Most students will be reading Sanskrit-in-IAST for the first time. Use Jamison & Brereton’s *Rigveda* translation for Ṛg 1.1; any standard for Īśa Upaniṣad.

Don’t push pronunciation. Push engagement with meaning.

Common misconceptions

1. **“NEP §4.6 invented IKS.”** False. The label is newer; the content is millennia old. NEP codified an existing tradition of teaching this material.

2. **“Sanskritists are anti-IKS.”** Most aren’t. Pollock, Cardona, Plofker — all do serious work on Indian intellectual history. The scholarly stance is “yes, this is real, AND we should describe it accurately.”
3. **“Indian math = Vedic math = Tirthaji.”** No. Indian mathematics has Bakhshali (~7th c. CE), Brahmagupta (7th c.), Bhāskara II (12th c.), Mādhava (14th c.). Tirthaji (1965) is a separate, more recent project.

Differentiation

- **Tier up:** Read Pollock’s *The Language of the Gods in the World of Men* chapter 1. Bring 1 question.
- **Tier down:** Focus on the canonical structure + 2 specific contributions. Skip the policy nuance.

Sensitivity

- The “IKS in policy” debate is *politically* loaded in contemporary India. Students from different political-cultural backgrounds will have strong priors. The classroom answer: each position has a strongest version, and the discipline is to articulate ALL of them fairly.

Paper oversight

- Strong topics name a specific claim + specific source + specific counter-argument.
- Weak topics generalise.
- The “honest limitation” criterion is the most important rubric item.

If you only have 6 days

- Day 1: Hook + vedic corpus
- Day 2: Six darśanas + math
- Day 3: Astronomy
- Day 4: Policy debate
- Day 5: Project session
- Day 6: Final + paper

Sources

Sources — Module 1 (Senior)

Primary

- **Ṛg Veda 1.1 (Agni hymn)** — Jamison & Brereton (2014), *The Rigveda*. Oxford.
- **Īśa Upaniṣad** — 18 verses; multiple translations. Standard: Olivelle (1996), *Upaniṣads*. Oxford.
- **Aryabhaṭīya (499 CE)** — Clark (1930) translation.

- **Bhāskara II, *Līlāvātī*** (1150 CE) — Patwardhan, Naimpally, Singh (2001).
- **Caraka Saṃhitā** — Sharma (1981).
- **Pāṇini, *Aṣṭādhyāyī*** — multiple editions; Cardona (1997) is the standard scholarly reference.

Scholarly secondary

- **Plofker, K. (2009).** *Mathematics in India*. Princeton.
- **Pollock, S. (2006).** *The Language of the Gods in the World of Men*. California.
- **Cardona, G. (1997).** *Pāṇini: His Work and Its Traditions*. Motilal Banarsidass.
- **Wujastyk, D. (2003).** *The Roots of Ayurveda*. Penguin.
- **Olivelle, P. (1996, ed.).** *Upaniṣads*. Oxford.
- **Hayashi, T. (1995).** *The Bakhshali Manuscript*. Egbert Forsten.

Comparative

- **Lloyd, G. E. R., & Sivin, N. (2002).** *The Way and the Word: Science and Medicine in Early China and Greece*. Yale.
- **Saliba, G. (2007).** *Islamic Science and the Making of the European Renaissance*. MIT.

Policy

- **Government of India (2020).** *National Education Policy 2020*. — §4.6 specifically.
- **Thapar, R. (2014).** *The Past as Present: Forging Contemporary Identities through History*. Aleph. (Critique of certain IKS framings.)
- **Kapoor, K. (various).** Writings on integrating Indian intellectual tradition into Indian education.

RAG cache

.rag-cache/day-01-intro.json — devotional-framing IKS chunks. Use verses for reference points; do NOT import the framing into the student-facing material.

Citation discipline

A strong Senior paper cites at least one primary, one scholarly, and one comparative source. Wikipedia is acceptable for orientation; never as a primary citation.

Day 1 — “IKS”: What Does the Term Refer To?

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students engage the boundary problem — what “IKS” denotes, what it doesn’t, and why the label itself is a recent construction.

Materials

- NEP 2020 §4.6 printed extract
- Whiteboard

Flow

Warm-up (5 min)

Provocation: “‘Indian Knowledge Systems’ — is that a coherent category, or a political slogan?”

Take 3 student answers. Don’t correct.

Core (25 min)

1. **The term “IKS” is recent.** Started gaining traction post-2000, codified by NEP 2020 §4.6. The *contents* — Vedic literature, Sāṅkhya philosophy, Pāṇini’s grammar, Bhāskara’s mathematics, Patañjali’s yoga, etc. — are millennia old.
2. **The boundary problem:**
 - Is “Buddhism” IKS? (Originated in India; spread to East Asia and beyond.)
 - Is “Sanskrit poetics” IKS? (Yes, but it’s also “Sanskrit literature.”)
 - Is “Mughal architecture” IKS? (Mixed Persianate-Indic; depends on definition.)
 - Is “Vedic mathematics” (Tirthaji 1965) IKS? (20th-c. synthesis with Vedic *label*.)
3. **Three honest framings to choose from** (write on board):

Framing	Strength	Weakness
“IKS = pre-modern Indian intellectual production”	Inclusive	Vague boundaries
“IKS = canonical Brahmanical knowledge systems”	Specific	Excludes Buddhist, Tantric, regional
“IKS = anything taught under the NEP §4.6 umbrella”	Operational	Political construction; varies by implementer

Each is defensible. Pick one knowingly, not by default.

4. **Read aloud NEP 2020 §4.6.** Discuss: what does the policy say IKS should include? What does it leave open?

Discussion (10 min)

Open question: “Which framing would you use in your paper? Why?”

Take 4 student positions.

Wrap-up (5 min)

- Preview Day 2: “Tomorrow — we read primary sources. *Ṛg Veda 1.1* and *Īśa Upaniṣad*.”

Differentiation

- **Tier up:** Read Pollock (2006) ch. 1 on Sanskrit as cosmopolitan language.
- **Tier down:** Focus on the three framings; don’t worry about the policy text yet.

Teacher notes

- Some students will immediately want to “defend Indian culture” against perceived critique. Reframe: precise definition IS defending it. Imprecise defence weakens.
- Some will want to dismiss the whole module as “Hindu nationalism.” Push: scholarly framing exists; engage it.

Citation(s) used

- Government of India (2020). NEP 2020 §4.6.
- vidya-karana RAG cache day-01-intro.json for background.

Day 2 — The Vedic Corpus: Close Reading

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students read *Ṛg Veda 1.1* and *Īśa Upaniṣad* opening in IAST with word-by-word gloss, and articulate the difference in tone between the two textual layers.

Materials

- Printed handout: *Ṛg Veda 1.1* (Sanskrit + IAST + Jamison-Brereton translation)
- Printed handout: *Īśa Upaniṣad* opening verse (Sanskrit + IAST + Olivelle translation)

Flow

Warm-up (5 min)

Recall: what are the four textual layers of the Veda? (Saṃhitā → Brāhmaṇa → Āraṇyaka → Upaniṣad.)

Reading 1: *Ṛg Veda 1.1* (15 min)

agnim īḷe purohitam yajñasya devam ṛtvijam hotāraṃ ratnadhātām “I praise Agni, the household priest, the divine officiant of the sacrifice, the invoker, who is the best bestower of treasures.”

- Word-by-word: *agnim* (Agni, acc.) + *īḷe* (I praise) + *purohitam* (household priest) + *yajñasya* (of the sacrifice, gen.) ...

- **Note:** this is a hymn of praise. The earliest layer of the Veda is ritual-centric.
- The intellectual contribution at this layer: precise oral transmission (across ~3000 years) generated the linguistic discipline that became Pāṇini.

Reading 2: Īśa Upaniṣad opening (15 min)

īśā vāsyam idam sarvaṁ yat kiñca jagatyāṁ jagat tena tyaktena bhuñjīthā mā grdhaḥ kasya svid dhanam “Whatever moves in this moving world is enveloped by the Lord. Enjoy what has been abandoned. Do not covet anyone’s wealth.”

- Word-by-word: *īśā* (by the Lord, inst.) + *vāsyam* (to be enveloped) + *idam sarvaṁ* (all this) ...
- **Note the move.** From ritual hymn (Saṃhitā) to ethical-philosophical reflection (Upaniṣad).
- The intellectual contribution at this layer: abstract metaphysical inquiry; the seed of Vedānta philosophy.

Discussion (5 min)

- Compare: the tone of the Agni hymn versus the Īśa Upaniṣad. What changed?
- The Vedic corpus contains BOTH — ritual and philosophy, hymn and abstract reflection. “Vedic” is a span, not a single voice.

Wrap-up (5 min)

- Preview Day 3: “Tomorrow — the six darśanas, the philosophical schools.”

Differentiation

- **Tier up:** Read 5 more verses of Īśa Upaniṣad. Identify one verse that surprised you.
- **Tier down:** Focus on the contrast between the two readings, not detailed Sanskrit.

Teacher notes

- Pronunciation perfection is NOT the goal. Engagement with meaning is.
- Some students will notice that the Upanishadic voice sounds “more modern philosophical.” That’s a real observation worth honouring.

Citation(s) used

- Jamison & Brereton (2014) for Ṛg Veda 1.1.
- Olivelle (1996) for Īśa Upaniṣad.

Day 3 — The Six Darśanas

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students name the six classical philosophical schools (darśanas) and the central question each addresses.

Materials

- Whiteboard
- Reference card with the six darśanas

Flow

Warm-up (5 min)

Recall: what's the difference between Vedāṅga and Upaveda?

Core (25 min)

The Six Darśanas (classical philosophical systems, codified ~early CE):

#	Sanskrit	Central question	Founder/text
1	<i>Nyāya</i>	Logic & epistemology — what counts as valid knowledge?	Gautama, <i>Nyāya Sūtras</i>
2	<i>Vaiśeṣika</i>	Atomism & metaphysics — what categories of being exist?	Kaṇāda, <i>Vaiśeṣika Sūtras</i>
3	<i>Sāṅkhya</i>	Cosmology — what are the constituents of reality?	Kapila / Īśvarakṛṣṇa, <i>Sāṅkhya Kārikā</i> (4th c. CE)
4	<i>Yoga</i>	Practice — how does one settle the mind?	Patañjali, <i>Yoga Sūtras</i>
5	<i>Mīmāṃsā</i> (Pūrva)	Hermeneutics — how do we interpret the Veda?	Jaimini, <i>Mīmāṃsā Sūtras</i>
6	<i>Vedānta</i> (Uttara Mīmāṃsā)	Metaphysics — what is the ultimate?	Bādarāyaṇa, <i>Brahma Sūtras</i> ; Śaṅkara, Rāmānuja, Madhva (later interpreters)

Note: these are often called the āstika schools (those that accept Vedic authority). The *nāstika* schools (Buddhism, Jainism, Cārvāka) developed in parallel — different intellectual traditions also “Indian,” not “IKS” by the conservative definition. Be explicit about this boundary choice.

Sāṅkhya and Yoga as a pair — closely related; Sāṅkhya provides the metaphysics (25 tattvas — see Module 2 Senior), Yoga provides the practice. You met Sāṅkhya in Module 2 (Panchabhuta) and will meet Yoga in Module 8.

Nyāya's contribution to logic. The Nyāya system developed sophisticated logical analysis — formal arguments (*anumāna*), categorical syllogisms, debate procedures (*tarka*). Influenced modern logic via early-20th-c. Indian logicians (especially Bimal Krishna Matilal's work in the 1960s).

Discussion (10 min)

- Which darśana would address each question?
 - “What does it mean to know something?” → Nyāya
 - “Why does my mind keep wandering?” → Yoga

- “How should we read the Veda?” → Mīmāṃsā
- “What is the relationship between the individual self and ultimate reality?” → Vedānta
- “What are matter’s smallest units?” → Vaiśeṣika

Wrap-up (5 min)

- Preview Day 4: “Tomorrow — Indian mathematics in specifics. Bakhshali, Bhāskara, Mādhava.”

Differentiation

- **Tier up:** Read Matilal, B. K. (1986). *Perception: An Essay on Classical Indian Theories of Knowledge*. Oxford.
- **Tier down:** Memorise the 6 names + central question of each.

Teacher notes

- The “āstika vs nāstika” distinction is important. Buddhism and Jainism are ENORMOUS Indian intellectual traditions excluded from a narrow “IKS” definition. Honest scholarship acknowledges this.

Citation(s) used

- Olivelle (1996, ed.) on the Upanishadic background.
- Matilal (1986) for Nyāya logic.
- Bryant (2009) for Yoga.

Day 4 — Indian Mathematics: Specifics

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students engage three specific Indian mathematical works at primary-source level, articulating what each established and what limits each had.

Materials

- Printed handout: one problem from Bhāskara II’s *Līlāvātī* in English translation
- Printed handout: statement of Mādhava’s series for π

Flow

Warm-up (5 min)

Recall: name 3 Indian mathematical contributions (from Day 6 of Middle Module 1, if students did it; or fresh recall).

Core (25 min)

1. **The Bakhshali manuscript** (~7th c. CE — fragments possibly earlier).
 - Birch-bark manuscript discovered in 1881.
 - Contains the earliest known use of zero as a number with a dot symbol.
 - Word problems on arithmetic, equations.
 - **Why it matters:** documents Indian arithmetic ~500 years before European adoption.
2. **Bhāskara II (1114–1185 CE)**. *Līlāvātī* and *Bījagaṇita*.
 - Arithmetic, algebra, geometry, mensuration.
 - Solved Pell's equation (the equation $x^2 - Ny^2 = 1$) centuries before Pell himself (1650s).
 - Introduced cycle methods for indeterminate equations.
 - Read aloud one *Līlāvātī* word problem (e.g. the swan problem).
3. **Mādhava of Saṅgamagrāma (~1340–1425 CE)**. Kerala school of mathematics.
 - Infinite series for π : $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$ (now sometimes called Madhava-Leibniz series).
 - Series for sine, cosine, arctan.
 - Came very close to calculus (~250 years before Newton and Leibniz).
 - **What he did NOT do:** did not develop the fundamental theorem of calculus formally; did not unify the series approach with a differential framework.

Discussion (10 min)

- “Madhava had infinite series 300 years before Leibniz. So why is the series still called ‘Leibniz series’ in most textbooks?”
- Possible answers: European mathematical literature was the dominant pipeline; Kerala school's work was not translated to European languages until much later; cite-attribution is conservative.
- The corrective: modern math history is increasingly using “Madhava-Leibniz series” — a real, slow correction.

Wrap-up (5 min)

- Preview Day 5: “Tomorrow — mid-module + paper topic commitment.”

Differentiation

- **Tier up:** Compute Mādhava's series numerically for 100 terms. How close to π ?
- **Tier down:** Focus on Bakhshali + Bhāskara names with dates.

Teacher notes

- The Madhava-Leibniz attribution is one of the most pedagogically useful examples of “real Indian contribution, slow Western recognition.”
- Don't oversell: Mādhava's work was discovered/recognised in the West via George Joseph and others in the late 20th c.

Citation(s) used

- Plofker (2009), Chapters 4–7.
- Hayashi (1995) on Bakhshali.
- George Joseph (2011). *The Crest of the Peacock: Non-European Roots of Mathematics*. Princeton.

Day 5 — Mid-Module + Paper Topic

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students commit to a paper topic, draft a thesis paragraph, and outline.

Materials

- Formative quiz (quizzes/formative.md Part B)
- Project brief

Flow

Quiz (20 min)

Closed-book. 8 questions.

Review + topic commitment (25 min)

- Brief class review of common errors (5 min).
- Each student writes:
 - Working title
 - One-paragraph thesis
 - 3 sources (≥ 1 primary, ≥ 1 scholarly, ≥ 1 comparative or policy)
- Quick buddy check (5 min).
- Submitted at door.

Homework

- Mark Day 6 reading. Bring questions.

Differentiation

- **Tier up:** Topic that engages the policy debate.
- **Tier down:** Topic on a single named achievement (e.g. “Mādhava’s series for π — what it established and what it did not”).

Day 6 — Indian Astronomy + The Yavana Exchange

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students articulate three specific Indian astronomical achievements AND the Greek-Indian intellectual exchange that complicates the “purely Indian” framing.

Materials

- Printed handout: one verse from Aryabhaṭīya in translation
- Printed extract from Varāhamihira’s *Pañca-siddhāntikā* (in translation)

Flow

Warm-up (5 min)

Recall: who was Aryabhata? When did he live?

Core (25 min)

1. **Aryabhata (~476–550 CE). *Aryabhaṭīya* (499 CE).**
 - Earth rotates on its axis.
 - Approximated π to 4 decimals.
 - Computed planetary periods.
 - Did NOT propose heliocentrism.
2. **Sūrya-siddhānta (early CE). Comprehensive astronomical text.**
 - Earth’s circumference computed within ~1% of modern value.
 - Detailed eclipse predictions.
3. **Varāhamihira (~6th c. CE). *Pañca-siddhāntikā*.**
 - **Compares 5 different astronomical systems** — including a “Romaka” (Greek-derived) system and a “Paulīśa” system possibly derived from Greek astronomer Paul of Alexandria.
 - **This is huge.** It documents that 6th-c. Indian astronomers were actively reading and engaging Greek astronomical work.
4. **The Yavana question.**
 - *Yavana* = “Greek” in Sanskrit. The Indo-Greek kingdoms (2nd c. BCE – 1st c. CE) created sustained cultural-intellectual exchange.
 - Greek astronomical methods influenced Indian astronomy.
 - Indian astronomical methods (Hindu numerals, trigonometric tables) influenced Arab and via Arab, European astronomy.
 - The “pure Indian” framing of IKS doesn’t survive contact with this evidence. The honest framing: Indian intellectual tradition is *partly indigenous, partly cosmopolitan*, in dynamic exchange.

Discussion (10 min)

- “Does the Yavana exchange weaken the case for celebrating Indian astronomical achievements?”
- The honest answer: no. *Synthesis* is also an achievement. Indian astronomers absorbed Greek methods AND developed their own (e.g. sine instead of chord, Hindu numerals).

Wrap-up (5 min)

- Preview Day 7: “Tomorrow — the contemporary policy debate. NEP §4.6.”

Differentiation

- **Tier up:** Read Pingree, D. (1996). “Astronomy in India” in *Astronomy Before the Telescope*.
- **Tier down:** Focus on the three achievements; the Yavana question is the depth layer.

Teacher notes

- The Yavana evidence is well-documented. Don’t shy from it. Honest history shows the Indian tradition as participant in a wider scholarly world.

Citation(s) used

- Aryabhata, *Aryabhaṭīya* (499 CE) — Clark (1930).
- Varāhamihira, *Pañca-siddhāntikā* — Neugebauer & Pingree (1971).
- Pingree (1996).

Day 7 — The Policy Debate: Should IKS Be Integrated into Mainstream Education?

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students articulate and defend a position on NEP 2020 §4.6 (IKS integration) with citation and counter-argument.

Materials

- NEP 2020 §4.6 printed extract
- Whiteboard

Flow

Warm-up (3 min)

Brief framing: today is structured debate. Three positions on the table.

Three positions (5 min)

Position	One-line claim	Scholarly defender (typical)
A — Integration	IKS should be taught alongside modern science in mainstream curriculum.	Kapil Kapoor; some NEP authors
B — Separation	IKS belongs in humanities/history class; modern science class should focus on the scientific method.	Romila Thapar; Ramachandra Guha (broadly)
C — Qualified integration	Historical context yes; equivalent scientific validity no.	Manjul Bhargava–style “math is universal”

Debate (25 min)

- 5 min: groups by position (self-select; balance).
- 8 min: group prep — 2 strongest arguments + 1 steel-manned counter to each of the other two.
- 10 min: structured debate (3 min per group + 1 min cross).
- 2 min: closing.

Oral assessment (during)

Teacher scores 5 random students on: position clarity, evidence use, counter-argument steel-manning, listening.

Wrap-up (12 min)

- 3 students share if their view shifted.
- *“You don’t have to commit tonight. Your paper position can be different from your debate position. That’s intellectually honest.”*

Differentiation

- **Tier up:** Take a position you DON’T hold. Hardest form of argument.
- **Tier down:** Use Day 5 reading response as your script.

Teacher notes

- This is the most politically charged session. Honour the discomfort.
- The honest framing: each position has a strongest version. The skill is articulating all three.

Citation(s) used

- NEP 2020 §4.6.

- Thapar (2014).
- Various Kapil Kapoor writings.

Day 8 — Project Session 1: Outline + Counter-Argument

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students refine thesis, complete outline, identify the strongest counter-argument.

Flow

Sprints (40 min)

- 10 min: refine thesis + outline
- 15 min: buddy critique — get the strongest counter-argument
- 15 min: incorporate, draft introduction

Wrap-up (5 min)

- Outline submitted.

Homework

- Draft 1.5–2 pages tonight.

Day 9 — Project Session 2: Draft

Module: What is IKS? (Senior) · **Time:** 45 min

Flow

Sprints (40 min)

- 20 min: silent drafting
- 15 min: buddy read + give one structural feedback
- 5 min: revise

Wrap-up (5 min)

- Submit polished draft of intro + 1 body section.

Homework

- Complete the paper at home tonight.

Day 10 — Summative + Paper Submission

Module: What is IKS? (Senior) · **Time:** 45 min

Flow

Quiz (35 min)

35 marks.

Submission (10 min)

- Hand in paper. Teacher spot-checks.
- Reflection slip.

Homework

- None.

Module 1 (Senior) — Formative Quizzes

Part A — Day 3 (8 min)

1.
 - (2) When did the term “IKS” gain currency as a unified category?
2.
 - (2) Name the 6 darśanas.
3.
 - (1) Which darśana provides the metaphysics underlying Yoga?
4.
 - (1) What distinguishes *āstika* from *nāstika* schools?
5.
 - (2) Name two Indian intellectual traditions excluded from a narrow “Brahmanical IKS” definition.

Key

1. Post-2000; codified by NEP 2020. 2. Nyāya, Vaiśeṣika, Sāṅkhya, Yoga, Mīmāṃsā, Vedānta. 3. Sāṅkhya. 4. Acceptance vs non-acceptance of Vedic authority. 5. Buddhism, Jainism (or Cārvāka).
-

Part B — Day 5 (15 min, 8 Q)

1.
 - (2) Translate word-by-word: *agnim īle purohitam* (Ṛg 1.1 opening).
2.
 - (2) What does the Īśa Upaniṣad opening verse say about ownership?

3. (1) The Bakhshali manuscript dates from approximately when?
4. (1) Who computed an infinite series for π in the 14th c. CE in Kerala?
5. (1) What did Aryabhata propose about the earth?
6. (1) What did he NOT propose?
7. (2) The Yavana exchange in astronomy — what does it show about “pure Indian” framings?
8. (3) Briefly state the three positions on the NEP IKS integration debate.

Key

1. agnim (Agni, acc.) īḥe (I praise) purohitam (household priest). 2. Do not covet what is not yours; whatever moves in this world is enveloped by the Lord. 3. ~7th c. CE. 4. Mādhava of Saṅgamagrāma. 5. The earth rotates on its axis. 6. Heliocentrism (sun-centred). 7. Indian astronomers actively absorbed Greek methods (Varāhamihira’s Pañca-siddhāntikā compares 5 systems including Greek-derived). “Pure Indian” overstates; the tradition was cosmopolitan. 8. (A) Integration — IKS in mainstream science class. (B) Separation — IKS in humanities/history. (C) Qualified integration — historical context yes, equivalent validity no.

Module 1 (Senior) — Summative Quiz

Time: 35 min · **Marks:** 35 · **Closed-book**

A — Multiple choice ($5 \times 1 = 5$)

1. The label “IKS” as a unified category gained traction:
 - a) ~1947 b) ~1970s c) Post-2000 d) Always existed
2. Which darśana focuses on logic and epistemology?
 - a) Sāṅkhya b) Nyāya c) Yoga d) Vedānta
3. The *Aṣṭādhyāyī* (~4th c. BCE) was written by:
 - a) Patañjali b) Pāṇini c) Bhāskara d) Mādhava
4. Mādhava’s infinite series for π was developed approximately:
 - a) 2nd c. BCE b) 500 CE c) 14th c. CE d) 17th c. CE
5. The Yavana exchange refers to:
 - a) Sanskrit-Tamil literary exchange b) Indo-Greek scholarly exchange
 - b) Arab-Indian medical exchange d) Chinese-Indian Buddhist exchange

B — Primary source (4)

- 6.

(4) Translate and briefly discuss: *īśā vāsyam idam sarvaṃ yat kiñca jagatyāṃ jagat*.

C — Canonical structure (5)

7. Name the 4 Vedas, 4 layers, 6 Vedangas, 4 Upavedas, 6 darśanas. (One per line; full marks if all correct.)

D — Specific contributions (6)

8.
(2) Bakhshali manuscript: date and significance.
9.
(2) Aryabhata's earth-rotation claim: what it does and does not establish.
10.
(2) Madhava-Leibniz attribution debate: why does it matter?

E — Policy (3)

11.
(3) State the three positions on IKS integration into mainstream education, with their typical scholarly defenders.

F — Open argument (12)

12.
(12) In 1.5–2 pages, argue your position on the following:
“The Indian intellectual tradition is best honoured by being described accurately, including its limits, rather than by uncritical celebration.”

Required: clear thesis, ≥1 source citation, steel-manned counter-argument.

Key (selected)

1c. 2b. 3b. 4c. 5b. 6. “Whatever moves in this moving world is enveloped by the Lord. Enjoy what has been abandoned. Do not covet anyone's wealth.” — A move from ritual to ethical/metaphysical reflection. 8. ~7th c. CE; earliest known use of zero as a number. 9. Establishes that earth rotates on its axis (correct, ~1000 years before European thinkers on this specific point). Does not establish heliocentrism (Copernicus, 1500s). 10. Shows real Indian achievement; shows Western citation conservatism; modern corrective is in progress.

Total: 35.

Homework

Homework — Module 1 (Senior)

Day	Task	Time
1 → 2	Read NEP §4.6. Note 2	30 min

Day	Task	Time
	ambiguities.	
2 → 3	Read 5 more verses of Īśa Upaniṣad in translation.	30 min
3 → 4	Read brief intro to one darśana (your choice).	30 min
4 → 5	Compute 100 terms of Madhava's π series; how close?	20 min
5 → 6	Submit paper plan: thesis + 3 sources.	30 min
6 → 7	Choose debate position; prepare 3-min argument + 1 steel-manned counter.	45 min
7 → 8	Draft 1 page of paper.	60 min
8 → 9	Continue drafting (2+ pages by tomorrow).	60 min
9 → 10	Complete and proofread.	60 min

Writing Prompts

Writing Prompts — Module 1 (Senior)

1. **(Day 1, reflection)** Is “IKS” a coherent category, a useful label, or a political construction? Defend in 1 page.
2. **(Day 2)** After reading Rg 1.1 + Īśa Upaniṣad — how would you describe the *intellectual* shift between the two textual layers? 1 page.
3. **(Day 4)** Argue: the Madhava-Leibniz attribution debate matters more for what it tells us about citation conventions than for mathematical priority. 1 page.
4. **(Day 6)** Defend or refute: “The Yavana exchange weakens the case for ‘purely Indian’ achievements in astronomy.” 1 page.
5. **(Day 7, paper potential)** Argue your position on NEP §4.6 IKS integration. Steel-man the opposing positions. 2 pages.
6. **(Cross-disciplinary)** Pick one Upaveda. Argue what its contemporary equivalent looks like in modern academic disciplines. 1 page.
7. **(Stretch)** Read the first chapter of Pollock's *Language of the Gods*. Engage his thesis. 2 pages.

Activity 01 — Primary Source Reading

Module: What is IKS? (Senior) · **Day:** 2 · **Time:** 30 min within Day 2

Goal

Students read Ṛg Veda 1.1 and Īśa Upaniṣad opening in IAST with word-by-word gloss.

Materials

- Printed handout (provided)

Instructions

Part 1: Ṛg Veda 1.1 (12 min)

Read silently. Then aloud (don't worry about perfect pronunciation).

Word-by-word with partner — match Sanskrit to gloss.

Answer: What is the speaker doing? What is the social context (priest, sacrifice)?

Part 2: Īśa Upaniṣad opening (12 min)

Same process. Read silently → aloud → word-by-word.

Answer: How is the tone different from Ṛg 1.1? What philosophical move does the verse make?

Discussion (6 min)

Pair share: “What surprised me about reading these in IAST.”

Differentiation

- **Tier up:** Read 5 more verses of Īśa Upaniṣad after class. Discuss tomorrow.
- **Tier down:** Skip the IAST pronunciation; focus on the meaning.

Project Brief

Project Brief — Module 1 (Senior)

Type: 5–6 page research paper · **Marks:** 65 · **Due:** Day 10

Topic options

Choose one or propose:

1. **Pick ONE specific Indian intellectual achievement** (Mādhava's series, Aryabhata's rotation, Pāṇini's grammar, Sushruta's surgery, the Nyāya theory of pramāṇas, etc.). Articulate what it established, what it didn't, and why it matters today.
2. **The policy question** — argue a position on NEP §4.6 IKS integration, steel-manning the other two positions.

3. **A comparative study** — pick a specific question (e.g. “the development of zero”) and compare the Indian, Greek, Chinese, and Arab traditions.
4. **The boundary question** — argue for a specific definition of “IKS.” Defend the boundary choice.

Required elements

- Clear thesis in opening paragraph.
- ≥ 1 primary source (verse, theorem, named text).
- ≥ 1 scholarly secondary source (Plofker, Pollock, Wujastyk, Cardona, Olivelle, Matilal).
- ≥ 1 comparative or policy source.
- A steel-manned counter-argument with response.
- Works-cited in MLA or Chicago.

Format

- 5–6 pages, double-spaced, 12pt.
- Title page with name, class, thesis.

Schedule

- Day 7: topic chosen.
- Day 8: outline.
- Day 9: draft.
- Day 10: paper submitted.

Rubric

See `rubric.md`.

Self-reflection

In 4–5 sentences: 1. Topic + why. 2. Most challenging aspect. 3. What you’d research deeper with 3 more days.

Rubric

Paper Rubric — Module 1 (Senior)

Paper marks: 65 (+ summative 35 → 100 total).

Criterion	4	3	2	1	Weight
Thesis clarity	Crystal-clear, specific	Clear	Vague	None	× 3

Criterion	4	3	2	1	Weight
Primary source	Quoted in IAST + translation + interpreted	Cited + interpreted	Vague	None	× 2
Scholarly source	Substantive engagement	Cited	Vague	None	× 2
Comparative / policy source	Engaged substantively	Cited	Vague	None	× 2
Counter-argument	Steel-manned + reasoned response	Counter presented	Acknowledged	Straw-manned	× 2
Honest limitation	Names what tradition did NOT do	Mentions	Briefly	Triumphalist	× 1
Originality	Original synthesis	Independent thinking	Mostly synthesis	Copied	× 1
Prose / structure	Polished	Clear	Issues	Hard to follow	× 1

Max raw: 56. Paper = (sum × weights) × 65/56.

Deductions

- Outside 5–6 pages: –5
- Missing works-cited: –5
- No IAST diacritics where Sanskrit appears: –3